

HR Employee Attrition — Analysis & Key Insights

Professional summary based on the supplied cleaned HR dataset

1. Executive Summary

This analysis examines employee attrition across departments, job roles, work patterns, tenure, compensation, satisfaction and other employee characteristics. The dataset contains 1,470 employee records and 28 attributes, with no missing values or duplicate employees.

Total Employees	Employees Left	Attrition Rate	Avg. Monthly Income
1,470	237	16.1%	6,502.93

2. Overall & Department Attrition

Overall attrition is 16.1% (237 of 1,470 employees). This company-wide rate is the reference point for comparing employee groups.

Department	Attrition Rate	Key context
Sales	20.6%	Highest department rate
Human Resources	19.0%	Small department; percentage can shift with few exits
Research & Development	13.8%	Lowest rate, but 133 of 237 exits because the department is largest

3. Key Attrition Signals

- **Overtime:** 30.5% attrition among employees working overtime versus 10.4% among those not working overtime.
- **Job role:** Attrition ranges from 39.8% for Sales Representatives to 2.5% for Research Directors.
- **Early tenure:** Employees with less than two years at the company show 34.9% attrition, compared with 8.1% for employees with more than ten years.
- **Work-life balance:** The lowest rating (1) has 31.2% attrition, while ratings 2–4 range between 14% and 18%.

4. Employee & Work Factors

Factor	Observed attrition pattern
Job Satisfaction	22.8% at level 1; 11.3% at level 4
Work-Life Balance	31.2% at level 1; 14–18% at levels 2–4
Job Level	26.3% at level 1; under 8% at the two most senior levels
Income	28.6% in lowest income band vs 5.6% in highest
Business Travel	24.9% frequent travel; 15.0% rare; 8.0% no travel
Performance Rating	16.1% rating 3 vs 16.4% rating 4 — almost identical

Job level, income, tenure and age describe overlapping employee groups, so they should not be interpreted as independent explanations.

5. Stayed vs Left — Employee Profile

Metric	Stayed (1,233)	Left (237)	Difference
Age	37.6	33.6	–4.0 years
Monthly income	6,833	4,787	–2,046
Years at company	7.4	5.1	–2.3 years
Years in current role	4.5	2.9	–1.6 years
Job level	2.15	1.64	–0.51
Job satisfaction	2.78	2.47	–0.31

Employees who left were, on average, younger, lower-paid, shorter-tenured and at lower job levels. Job satisfaction was also lower, but the gap was smaller.

6. Key Takeaways

- Overall attrition is **16.1%**, with 237 employees leaving out of 1,470.
- The clearest group-level signal is **overtime**: 30.5% versus 10.4%.
- Attrition is concentrated among **early-tenure employees**: 34.9% within the first two years versus 8.1% after ten years.
- **Junior and lower-paid employees** show higher attrition, overlapping with younger age and shorter tenure.
- **Job role** reveals much larger differences than department-level reporting.
- Performance ratings are almost identical between ratings 3 and 4, so attrition is not simply concentrated among lower-rated performers.

7. Methodology & Limitations

Attrition rate = employees with Attrition = Yes divided by all employees in the relevant group. Constant fields such as EmployeeCount, Over18 and StandardHours were excluded. All figures are calculated from the supplied dataset.

Important: These findings describe observed differences in historical employee records. The dataset does not include exit dates, exit reasons, or whether departures were voluntary, so the results identify groups with higher or lower attrition rather than proving why individual employees left.